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Efficacy and safety of creatine supplementation in patients with heart failure and reduced ejection fraction: a pilot study



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ABSTRACT

Introduction and objectives: Heart failure (HF) poses a significant global health burden. Patients with HF experience reduced exercise tolerance, greatly affecting their quality of life. While creatine monohydrate (CM) improves functional capacity in healthy individuals, its effects in HF patients remain unexplored.

Methods: We conducted an exploratory, prospective, single center, open-label study evaluating the efficacy and safety of 5 g/day CM during 3 months in patients with HF and reduced ejection fraction. Outcomes included functional capacity, health-related quality of life (HRQoL), echocardiographic and laboratory parameters at 3-months and HF decompensations and mortality at 1-year.

Results: A total of 43 patients participated (60.73 ± 11.73 years; 30.43% females). The intervention was safe. Creatinine increased and glomerular filtration rate declined at 3 months but normalized after discontinuing CM. We observed a statistically significant increase in the 6-minute walk test (48.69 ± 32.76; $P = .005$), a reduction in Borg scale exertion (−1.57 ± −1.73 points; $P < .01$), and decreased heart rate increment during exercise (−2.43 ± −8.66 bpm; $P = .04$). HRQoL improved significantly (4.08 ± 12.29 points; $P = .03$). No changes were found in echocardiographic parameters, HF decompensations, or mortality.

Conclusions: CM supplementation in patients with HF and reduced ejection fraction is safe and could improve functional capacity and HRQoL. Further research is warranted to confirm our results.

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Abbreviations: 6MWT: 6-minute walk test; CM: creatine monohydrate; HFREF: heart failure with reduced left ventricular ejection fraction; HRQoL: health related quality of life; KCCQ: Kansas city Cardiomyopathy Questionnaire; MCID: minimal clinically important difference.

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Eficacia y seguridad de la suplementación con creatina en pacientes con insuficiencia cardíaca y fracción de eyección reducida: un estudio piloto

R E S U M E N

Palabras clave:

Insuficiencia cardíaca
Creatina
Calidad de vida
Capacidad funcional
Test de la marcha

Introducción y objetivos: La insuficiencia cardíaca (IC) constituye una pandemia que impacta en la salud pública global. Los pacientes con IC experimentan una reducción en la tolerancia al ejercicio que afecta su calidad de vida. El monohidrato de creatina (MC) mejora la capacidad funcional en individuos sanos, pero sus efectos en pacientes con IC no han sido explorados. El objetivo del estudio es evaluar la eficacia y seguridad del MC en la IC con fracción de eyección reducida.

Métodos: Se realizó un estudio exploratorio, prospectivo, unicéntrico y abierto, con MC 5 g/día durante 3 meses. Las variables de resultado fueron capacidad funcional, calidad de vida relacionada con la salud (CVRS) parámetros ecocardiográficos y de laboratorio a los 3 meses, así como reagudizaciones de IC y mortalidad al año.

Resultados: Participaron 43 pacientes ($60,73 \pm 11,73$ años; 30,43% mujeres). La intervención fue segura. La creatinina aumentó y el filtrado glomerular disminuyó a los 3 meses, normalizándose al suspender el MC. Observamos un aumento estadísticamente significativo en el test de la marcha de seis minutos ($48,69 \pm 32,76$; $p = 0,005$), una reducción en la escala de Borg ($-1,57 \pm -1,73$ puntos; $p < 0,01$) y una disminución del incremento de frecuencia cardíaca durante el ejercicio ($-2,43 \pm -8,66$ lpm; $p = 0,04$). La CVRS mejoró significativamente ($4,08 \pm 12,29$ puntos; $p = 0,03$). No se encontraron diferencias en parámetros ecocardiográficos, reagudizaciones o mortalidad.

Conclusiones: El MC en pacientes con IC y fracción de eyección reducida es seguro y podría mejorar la capacidad funcional y CVRS. Se necesitan más estudios para confirmar nuestros resultados.

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Introduction

Heart failure (HF) is a global health pandemic that affects 26 million people worldwide, and is increasing rapidly due to the aging population.¹ In 2021, HF was defined as a clinical syndrome with symptoms (e.g. dyspnea, and fatigue) and/or signs (e.g. pulmonary crackles and peripheral edema) caused by a structural and/or functional cardiac abnormality and corroborated by elevated natriuretic peptide levels and/or objective evidence of pulmonary or systemic congestion.^{2,3} Exercise intolerance, dyspnea, and fatigue are hallmark features of HF. The pathophysiological mechanisms underlying diminished functional capacity in HF are multifactorial, including both central and peripheral mechanisms such as abnormal skeletal muscle metabolism, muscle atrophy and a switch from slow type I to fast type II myofibrils.^{4,5} Furthermore, sarcopenia, characterized by reduced muscle mass and strength, affects up to 30–50% of patients with HF and reduced ejection fraction (HFrEF),⁶ contributing to increased morbidity.^{7–9}

Given the high prevalence of HF and its profound impact on both functional capacity and health related quality of life (HRQoL), comprehensive interventions aimed at alleviating symptoms are crucial. Emerging evidence suggests that creatine may offer potential benefits for patients with HF. Creatine is a natural molecule comprised of 3 amino acids (methionine, arginine and glycine) that is primarily stored in skeletal muscle but also exists in the myocardium.¹⁰ It plays a pivotal role in

energy metabolism by enabling rapid adenosine triphosphate (ATP) resynthesis, thereby enhancing muscular performance (Fig. 1).^{11,12} Consequently, creatine stands as the most widely used and studied supplement in athletes, demonstrating a broad spectrum of benefits, such as enhanced muscle strength and recovery, muscle mass gain, improved glycogen synthesis, and increased endurance.^{13,14} Some studies also suggest potential benefits for bone structure, cognitive functions, and even mood.¹⁵

Creatine preparations range from 5 g/day to 30 g/day, with studies showing that doses can vary widely to achieve effective intramuscular concentrations. However, the most commonly used and well-researched form is creatine monohydrate (CM) at a dose of 5 g/day.^{13,16} In addition to the benefits it provides, CM has demonstrated to be safe, with the most common side effect being a slight increase in weight, attributed to enhanced muscle glycogen storage.^{9,17,13} A mild increase in serum creatinine (Cr) levels is observed during supplementation, but this rise is transient and does not affect other renal function parameters.^{18,19}

In recent years, the use of CM has extended beyond the realm of athletic performance and has been tested in neurological disorders such as Parkinson's disease, mitochondrial encephalomyopathies and amyotrophic lateral sclerosis, suggesting potential improvement in muscle endurance, strength, exercise capacity and HRQoL.^{20–23} This application has sparked interest in exploring CM supplementation in HF. The benefits are 2-fold. Firstly, it could improve car-

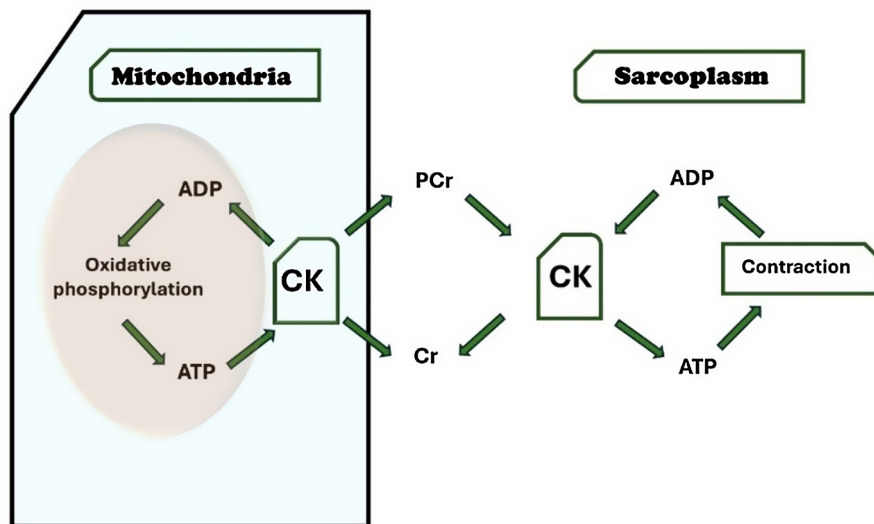


Fig. 1 – Metabolic pathway: creatine, phosphocreatine, and ATP cycling. ADP, adenosine diphosphate; ATP, adenosine triphosphate; CK, creatine kinase; Cr, creatine; PCr, phosphocreatine.

diac function, and potentially slow myocardial deterioration by replenishing intramyocardial creatine, phosphocreatine, and ATP levels. Secondly, it could enhance skeletal muscle performance helping to prevent sarcopenia. Experimental studies have demonstrated increased myocardial performance with in vitro supplementation,²⁴ but in vivo evidence remains limited. Previous research suggests CM in patients with HF improves functional capacity and muscle strength, while maintaining safety.^{25–28} Nonetheless, these studies have small sample sizes and heterogeneous protocols and dosages.

Therefore, the aim of the current study was to investigate the impact of oral CM on functional capacity measured by the 6-minute walk test (6MWT) and HRQoL assessed by the Kansas City Cardiomyopathy Questionnaire (KCCQ). Additionally, we sought to evaluate the safety of the intervention and its impact on blood analysis and echocardiographic parameters, as well as long-term outcomes, including HF decompensations and cardiovascular (CV) mortality at 1 year. We hypothesized that oral CM would enhance functional capacity and HRQoL in HF patients and have no effect on echocardiographic, blood analysis parameters, HF decompensations or CV mortality.

Methods

Patients

Participants were recruited from the HF clinic of the General University Hospital of Santa Lucía, Cartagena, from November to December of 2021. Inclusion criteria were: (a) male and female patients diagnosed with HFrEF (left ventricular ejection fraction [LVEF] < 40%) of any etiology; (b) aged ≥ 18 years; (c) who were clinically stable for at least 3 months prior to inclusion; clinical stability was defined by the absence of symptoms, changes in medical treatment, hospital admis-

sions or visits to the emergency department due to HF; (d) New York Heart Association (NYHA) functional class I–III.; and (e) no physical limitations to exercise. Exclusion criteria were: (a) age > 80 years; (b) NYHA IV; (c) chronic kidney disease (CKD) stage > G3b; (d) physical and/or psychological limitations to exercise.

Study design

Our study was a single-center, prospective, non-controlled pilot investigation. The protocol was approved by the hospital's Ethics Committee (Ref: E.O. 2021-69 Suplementacion-CreatinaIC-FEVIR; date of approval: 26/10/2021). Informed written consent was obtained from all participants. The study protocol is shown in Fig. 2. During the initial visit, we collected detailed medical histories and conducted physical examination, gathering anthropometric data (i.e., weight, height, and body mass index [BMI]). We also recorded self-reported physical activity levels and demographic information from electronic databases. Baseline assessments included the 6MWT, blood analyses, echocardiography and HRQoL evaluation using the KCCQ. Blood analyses included a complete blood count, biochemistry, renal function (Cr and estimated glomerular filtration rate [eGFR] calculated with the CKD-EPI formula), and biomarkers (N-terminal pro-B-type natriuretic peptide, [NT-ProBNP] and Cancer Antigen 125 [CA125]). Echocardiographic measurements encompassed LVEF, telediastolic and telesystolic left ventricular diameters. At the 6-week midpoint, we repeated blood analyses as a safety measure. After 3 months, we conducted follow-up evaluations, including 6MWT, KCCQ, blood analyses and echocardiography. Blood analysis was repeated at 6 months to assess safety. We meticulously recorded emergency visits, hospitalisations related to HF, and CV mortality 1-year post-intervention using electronic databases.

Primary outcomes were exercise capacity measured by the 6MWT and HRQoL using the KCCQ. Secondary outcomes

Efficacy and safety of creatine supplementation in patients with heart failure and reduced ejection fraction

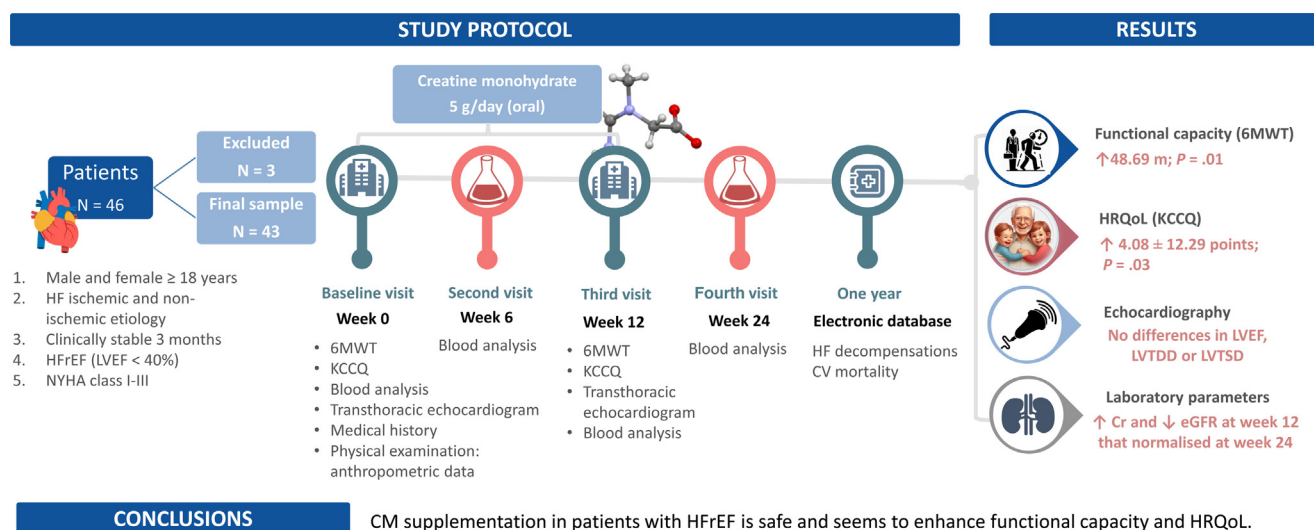


Fig. 2 – Central illustration. Study protocol on the efficacy and safety of creatine monohydrate supplementation in patients with heart failure and reduced ejection fraction. 6MWT, 6-minute walk test; CM, creatine monohydrate; Cr, creatinine; CV, cardiovascular; eGFR, estimated glomerular filtration rate; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; HRQoL, health-related quality of life; KCCQ, Kansas City cardiomyopathy questionnaire; LVEF, left ventricular ejection fraction; LVSD, left ventricular systolic diameter; LVTTD, left ventricular telediastolic diameter; NYHA, New York Heart Association.

included laboratory and echocardiographic parameters, as well as HF decompensations and CV mortality.

Creatine supplementation

During the 3-month intervention period, patients received 5 g of oral Creapure-certified CM daily. Creatine was provided in sealed unit doses. We selected a dosage of 5 g/day because it has proven effective in attaining adequate intramuscular concentrations and is commonly used in both athletic and non-athletic settings.^{13,15,17,29} While higher doses are safe, they do not enhance effectiveness due to a saturation limit in intramuscular creatine storage.²⁹ Adherence was monitored by counting returned containers at the twelve-week check-up visit. All other medication remained unchanged during the intervention, and patients did not follow a structured exercise regime since recruitment began after the completion of their cardiac rehabilitation program.

Statistical analyses

Descriptive statistics were calculated for categorical variables using frequencies and proportions, while mean differences (MD) and standard deviations (SD) were used for continuous variables. Paired t-tests were conducted to compare baseline and follow-up measurements. McNemar's test was applied to evaluate categorical outcomes. A P value < .05 was considered statistically significant. Data analysis was performed using Stata version 14.2 (StataCorp, 2015, United States).

Results

Patient characteristics

A total of 46 patients met the inclusion criteria and were recruited. Nonetheless, 3 patients were excluded: 1 due to loss of follow up and 2 due to mortality, including one from COVID-19 infection and another from ventricular tachycardia. Therefore, the final study population comprised 43 individuals (12 women; 27.91%), with a mean age of 59.74 ± 11.40 years. Demographic and clinical characteristics of these patients are reported in Table 1.

Outcomes measured

Table 2 summarizes the outcomes measured at baseline and after the 3-month intervention. Fig. 2 provides an overview of the study's design, participants and results.

Safety and long term outcomes

No self-reported adverse events occurred during the intervention or follow-up period. In the 12 months following the intervention, 7 patients (16%) experienced HF decompensation. In all cases, the decompensation was triggered by factors unrelated to the intervention, such as infections, arrhythmias, or treatment discontinuation.

Functional capacity (6MWT)

Our study found a statistically significant improvement in the distance covered in the 6MWT (48.69 ± 32.76 m; $P = .01$; Fig. 3A) post-intervention compared to baseline values. Additionally, there was a statistically significant decrease in patients' perceived exertion, measured using the Borg scale (-1.57 ± -1.73 points; $P < .01$; Fig. 3B), along with a significant reduction in HR increment during exertion (-2.43 ± -8.66 bpm; $P = .04$; Fig. 3C).

Health related quality of life

After the intervention, there was a statistically significant improvement in HRQoL assessed with the KCCQ total score compared to baseline (4.08 ± 12.29 points; $P = .03$; Fig. 4); When

breaking down the results by KCCQ domains, we only observed significant increases in the 5th (social limitations) (6.44 ± 17.34 points; $P = .01$) and 7th domain (self-efficacy) (6.04 ± 14.49 points; $P < .01$). While improvements were noted in other KCCQ domains, these changes did not reach statistical significance.

Echocardiography

No significant differences were observed post-intervention in echocardiographic parameters, including LVEF assessed by Simpson's biplane method ($MD \pm SD = 0.47 \pm 3.08$; $P = .46$), as well as left ventricular size assessed by telediastolic ($MD \pm SD = 0.42 \pm 1.99$ mm; $P = .09$) and telesystolic diameters ($MD \pm SD = -0.9 \pm -2.3$ mm; $P = .40$).

Table 1 – Baseline characteristics of the population.

Variable	N = 43	Variable	N = 43
Sex (male/female)	31 (72.09)/ 12 (27.91)	Smoker	
		Never	8 (18.60)
		Active	4 (9.30)
		Previous	31 (72.09)
Age, years	59.74 ± 11.40	Alcoholism	
		Never	32 (74.42)
		Active	2 (4.65)
		Previous	9 (20.93)
BMI, kg/m ²	30.56 ± 5.10	LVEF, %	33 ± 5.04
Hypertension	30 (69.77)	NYHA class	
		I	20 (46.51)
		II	19 (44.19)
		III	4 (9.30)
Diabetes	20 (46.51)	AF or Flutter	11 (25.58)
Dyslipidemia	30 (69.77)	Cardiac implantable devices	
		PM	1 (2.33)
		ICD	22 (51.16)
		CRT	8 (18.60)
CKD (eGFR < 60 mL/min/m ²)	24 (55.81)	Physical activity	
		None	11 (25.58)
		Walking	26 (60.47)
		Cycling	7 (16.28)
Cr (mg/dL)	1.11 ± 0.25	eGFR (mL/min/1.73 m ²)	70.58 ± 16.87
Ischemic cardiomyopathy	21 (48.84)	Charlson score	
		0–1 pts	3 (6.98)
		2 pts	3 (6.98)
		≥3 pts	37 (86.05)
Other etiologies		HF treatment	
Alcoholic	4 (9.30)	Beta-blockers	43 (100)
Chemotherapy	2 (4.65)	ACEI/ARB	10 (23.26)
Miocarditis	2 (4.65)	ARNI	33 (76.74)
Idiopathic	14 (32.56)	MRA	40 (93.02)
		iSGLT2	26 (60.47)
		Loop diuretics	25 (58.14)

ACEI, angiotensin-converting enzyme inhibitor; AF, atrial fibrillation; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; BMI, body mass index; CKD, chronic kidney disease; Cr, creatinine; CRT, cardiac resynchronization therapy; eGFR, estimated glomerular filtration rate; HF, heart failure; ICD, implantable cardioverter defibrillator; iSGLT2, sodium-glucose cotransporter-2 inhibitor; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NYHA, New York Heart Association; PM, pacemaker; SD, standard deviation. Data are expressed as no. (%) or mean $\pm$ standard deviation.

Table 2 – Outcomes measured at baseline and after 3 months of oral creatine monohydrate supplementation.

Variable	Baseline mean ± SD	Post-intervention (week 12) mean ± SD	Δ Mean ± SD	P (95%CI)
Functional capacity (6MWT)				
6MWT (m)	349.40 ± 72.89	398.10 ± 72.38	48.69 ± 32.76	.01 (38.48–58.90)
ΔHR exertion (bpm)	11.64 ± 8.11	9.21 ± 5.84	–2.43 ± –8.66	.04 (–5.13–0.27)
Borg scale (points)	5.21 ± 2.18	3.64 ± 2.12	–1.57 ± –1.73	.01 (–2.11 to –1.03)
Health related quality of life (KCCQ)				
Total score (points)	75.06 ± 14.12	79.14 ± 14.37	4.08 ± 12.29	.03 (–7.86 to –0.29)
1st domain (points) Symptom frequency	81.14 ± 17.75	77 ± 17.08	5.11 ± 15.26	.33 (–9.81 to –0.41)
2nd domain (points) Symptom burden	77 ± 17.08	78.68 ± 17.18	1.67 ± 16.73	.51 (–6.82 to 3.47)
3rd domain (points) Symptom stability	88.83 ± 25.18	94.41 ± 17.08	5.58 ± 19.18	.06 (–11.48 to 0.32)
4th domain (points) Physical limitation	72.72 ± 18.17	77.97 ± 19.62	5.25 ± 17.79	.06 (–10.73 to 0.22)
5th domain (points) Social limitation	75.41 ± 19.84	81.86 ± 17.43	6.44 ± 17.34	.01 (–11.78 to –1.10)
6th domain (points) Quality of life	68.68 ± 19.92	71.62 ± 19.89	2.94 ± 18.41	.30 (–8.61 to 2.72)
7th domain (points) Self-efficacy	77.67 ± 17.84	83.72 ± 17.86	6.04 ± 14.49	<.01 (–10.50 to –1.58)
Total symptom score (points)	79.37 ± 17.09	83.01 ± 16.38	3.64 ± 14.77	.11 (–0.90 to 8.19)
Clinical summary score (points)	78.53 ± 16.43	82.38 ± 16.37	3.84 ± 14.30	.08 (–6.99 to 0.60)
Overall summary score (points)	75.81 ± 16	79.65 ± 16.37	3.83 ± 13.50	.06 (–7.99 to 0.31)
Transthoracic echocardiography parameters				
LVEF 4C (%)	32.93 ± 5.11	32.86 ± 4.49	0.07 ± 4.43	.46 (–1.43 to 1.29)
LVEF 2C (%)	33.02 ± 6.98	33.44 ± 6.67	0.42 ± 6.61	.31 (–1.30 to 2.14)
LVEF bp (%)	33 ± 5.04	33.05 ± 4.63	0.47 ± 3.08	.46 (–0.90 to 0.99)
LVTDD (mm)	63.65 ± 8.35	64.07 ± 8.24	0.42 ± 1.99	.09 (–0.19 to 1.03)
LVTSD (mm)	58.02 ± 7.91	57.93 ± 7.73	–0.9 ± –2.3	.40 (–0.80 to 0.62)

6MWT, 6-minute walk test; 95%CI, 95% confidence interval, Cr, creatinine; CV, cardiovascular; eGFR, estimated glomerular filtration rate; HF, heart failure; KCCQ, Kansas City Cardiomyopathy Questionnaire; LVEF bp, left ventricular ejection fraction, biplane Simpson measurement; LVEF 4C, left ventricular ejection fraction, 4 chamber Simpson measurement; LVEF 2C, left ventricular ejection fraction, 2 chamber Simpson measurement; LVTDD, left ventricular telediastolic diameter; LVTSD, left ventricular telesystolic diameter; ΔHR, heart rate increment.

Blood analysis

Table 3 summarizes blood analysis parameters measured at baseline and at weeks 6, 12 and 24. Regarding serum Cr concentrations, a statistically significant increase was observed at week 12 (0.13 ± 0.24 mg/dL; $P = .01$), which normalized during follow-up at week 24 and even showed improvement compared to baseline values (0.02 ± 0.24 mg/dL; $P = .01$; Fig. 5). Similar trends were observed with eGFR, which showed a decrease at week 12 (-6.23 ± 11.81 mL/min; $P = .01$) followed by an improvement at week 24 compared to baseline (6.27 ± 11.97 mL/min; $P = .01$; Fig. 5). Plasma concentrations of CA125 and NT-proBNP did not exhibit statistically significant variations during the intervention period or at week 24 compared to baseline values.

Discussion

The current study aimed to investigate whether a short period of CM supplementation (5 g/day for 3 months), enhances exercise capacity, HRQoL, echocardiographic and laboratory parameters, in patients with HF. We also sought to evaluate the safety of this intervention.

In accordance with our initial hypothesis, CM supplementation proved safe for HF patients. No adverse outcomes were reported during supplementation, and we did not observe any significant increase in HF decompensations during the intervention or at 1-year follow-up. The rate of HF decompensation was 16% at 1-year, comparable to the annual rate reported

in the ESC-HF-LT registry for patients with HFrEF.³⁰ Additionally, there was no deterioration in blood analysis markers. The transient increase in serum Cr levels observed at week 12 is attributed to the rapid breakdown of CM into Cr, subsequently excreted by the kidneys.³¹ The apparent decline in eGFR during supplementation is a consequence of using serum Cr in eGFR formulas.³² Both Cr and eGFR normalized upon cessation of CM at week 24. These findings do not indicate kidney toxicity but rather a normal consequence of CM supplementation. Numerous studies have consistently supported the renal safety profile of CM across both short- and long-term regimens.^{18,19,33} In studies estimating GFR with alternative methods like cystatin C, no changes in renal function were observed.¹⁸

Regarding the efficacy of the intervention, we found that CM improves exercise capacity, as measured by the distance covered in the 6MWT. This result was statistically significant and clinically relevant, exceeding the minimal clinically important difference (MCID) of 35–37 m reported for HF patients.³⁰ Compared to other treatments for patients with HFrEF, CM obtained higher improvements in the 6MWT (48.69 ± 32.76 m). For instance, the DETERMINE trial showed that Dapagliflozin 10 mg/day during 4 months, improved the 6MWT by 20 m (interquartile range (IQR), [–2 to 42]).³⁴ Similarly, a pilot study of 58 patients with HFrEF treated with Sacubitril/Valsartan showed an improvement in the 6MWT of 41.8 m (IQR, [33.4–50.2]; $P < .001$) compared to baseline.³⁵ Conversely, a comprehensive review and meta-analysis revealed that cardiac rehabilitation programs enhanced the 6MWT by an average of 60.43 m.³⁶ These results are important because

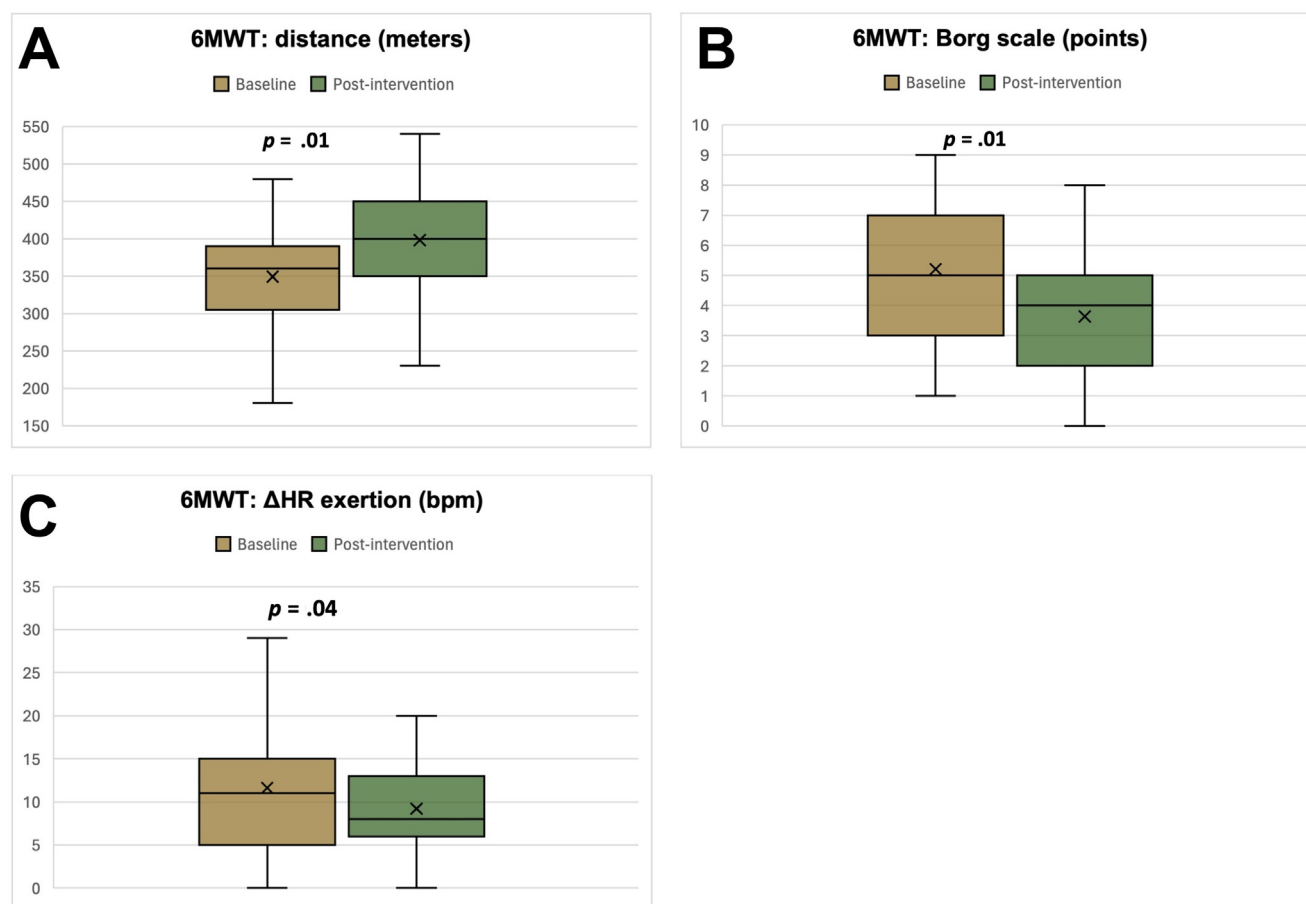


Fig. 3 – Changes in the distance covered in the 6-minute walk test (A), Borg scale of perceived exertion scale (B), and the heart rate increment with exercise (C), post-intervention compared to baseline values. 6MWT, 6-minute walk test; bpm, beats per minute; ΔHR, delta heart rate.

Table 3 – Laboratory parameters measured at baseline and at weeks 6, 12 and 24.

Variables	Baseline Mean ± SD	Change week 0–6		Change week 0–12		Change week 0–24	
		Mean ± SD	P 95%CI	Mean ± SD	P 95%CI	Mean ± SD	P 95%CI
Creatinine (mg/dL)	1.11 ± 0.25	−0.02 ± 0.17	.21 (−0.07 to 0.03)	0.13 ± 0.24	.01 (0.05–0.20)	−0.09 ± 0.18	.01 (−0.14 to −0.03)
eGFR (mL/min/1.73m ²)	70.58 ± 16.87	1.98 ± 10.60	.11 (−1.29 to 5.24)	−6.23 ± 11.81	.01 (−9.87 to 2.60)	6.27 ± 11.97	.01 (2.59–9.96)
CA125 (U/mL)	12.15 ± 10.16	−0.7 ± 4.39	.15 (−2.07 to 0.67)	1.15 ± 14.80	.24 (−3.00 to 6.11)	2.70 ± 17.76	.16 (−2.76 to 8.17)
NT-ProBNP (pg/mL)	729.40 ± 698.10	−157.14 ± 287.62	.01 (−245.66 to 68.62)	93.19 ± 689.72	.19 (−119.08 to 305.45)	−150.07 ± 890.32	.14 (−123.93 to 424.07)

CA125, carcinoembryonic antigen 125; 95%CI, 95% confidence interval; eGFR, estimated glomerular filtration rate; NT-ProBNP, N-terminal pro-B-type natriuretic peptide; SD, standard deviation.

exercise capacity is an independent predictor of mortality and CV events in patients with HF.^{37,38} Notably, these gains were achieved with CM alone, without a structured exercise regimen. Previous evidence suggests that muscle creatine uptake is enhanced when submaximal exercise is performed during supplementation, implying that combining CM with exercise could yield even greater benefits.^{39,40} This contrasts with previous studies that found no differences in functional capacity with CM in patients with HF compared to placebo.^{25,26,28} Carvalho et al. studied the effect of 5 g/day of CM during 6 months in 33 male patients with HF and found no significant differences in functional capacity as measured by the 6MWT and VO₂ peak.²⁵ They suggested that the absence of an exercise

routing during CM supplementation might account for the lack of improvement. Similarly, Kuethe et al. found no significant improvements in the 6MWT, VO₂ peak or VO₂ at the anaerobic threshold with 20 g/day of CM for 6 months in HF patients with reduced exercise capacity.²⁸ Comparing these results with our own is challenging due to the lack of a control group, a limitation we acknowledge.

Besides improvements in the 6MWT, we also found a reduction in patients' perceived exertion. Post-intervention, Borg scale scores decreased by a mean of 1.56% compared to baseline, a result considered clinically meaningful as it exceeds the 1-point threshold recognized as the MCID in patients with HF.⁴¹ There was also a significant reduction in HR incre-

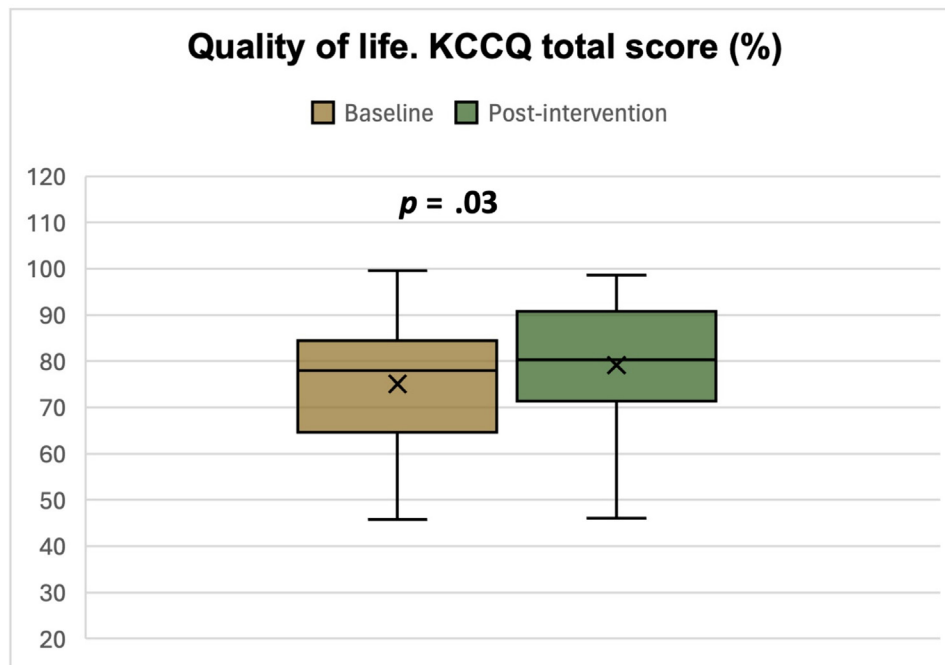


Fig. 4 – Changes in quality of life measured by the Kansas City Cardiomyopathy Questionnaire (KCCQ) total score post-intervention compared to baseline values.

ment during exercise by 2.43 bpm, which could also indicate enhanced exercise tolerance. These findings show both objective and subjective improvements in exercise tolerance, a key goal in HF management.

Additionally, CM enhanced HRQoL in patients with HF, as measured by the KCCQ. The improvement we observed in the total score post-intervention (4.08 ± 12.29 points) was statistically significant, but did not reach the MCID of 5 points necessary for clinical relevance.⁴² Regarding the different KCCQ domains, all of them increased post-intervention, although only 2 reached statistical significance. The improvement in KCCQ total symptom score in our study (3.64 ± 14.77 points), which is calculated by combining the first and second domain, exceeded that of other interventions in HFrEF. The DETERMINE trial reported an increase in the KCCQ total symptoms score of 2.1 points (IQR, [−4.2 to 14.6], in the Dapagliflozin group after 4 months.³⁴ Improving HRQoL is a major treatment goal for HF patients due to the condition's impact on morbidity.² Individuals with HF suffer from substantial symptoms, physical limitations and impaired HRQoL compared to healthy individuals and patients with other chronic diseases.⁴³ Furthermore, HRQoL has demonstrated to be a strong and independent predictor of mortality and HF hospitalisations.⁴⁴ These results contrast with previous research. Kuethe et al. found no difference in HRQoL, measured with the Minnesota living with HF questionnaire, in patients with HF supplemented with 20 g/day CM for 6 months compared to placebo.²⁸ Similarly, Cornelissen et al. reported that 9 g/day of CM for 3 months did not improve HRQoL, measured with the SF-36 and McNew Heart disease questionnaires.²⁶

Regarding cardiac function parameters, previous evidence indicates that diseased hearts, both human and animal models, exhibit at least a 20% reduction in total creatine content compared to healthy myocardium.^{3,7} In fact, a lower phosphocreatine-to-ATP ratio is a stronger predictor of CV mortality in patients with dilated cardiomyopathy than LVEF.⁴⁵ This suggests that CM might improve cardiac function. However, we found no changes in systolic function measured by echocardiography, aligning with previous studies.⁴⁶ For instance, Kuethe et al. found no increase in LVEF or a decrease in left ventricular telediastolic diameter with 20 g/day of CM during 6 months in HF patients.²⁸ The need for a larger patient population to detect significant changes in LVEF could explain these findings due to substantial intra- and interobserver variability.^{4,5}

Strengths and limitations

Our study addresses an important gap in the literature, as existing research on CM supplementation in HFrEF patients is outdated. Few studies have explored the effects of CM alongside current treatment regimens. The 3-month supplementation period in our study, combined with long-term safety evaluations post-discontinuation, provides valuable insights into the benefits and safety of creatine. However, some limitations should be noted. First, the study's small sample size limits the generalisability of our findings. Secondly, although VO₂ peak is the gold standard for assessing functional capacity in clinical settings, we were unable to measure it due to the unavailability of cardiopulmonary exercise testing at our center.³² Nonetheless, the 6MWT is a widely used

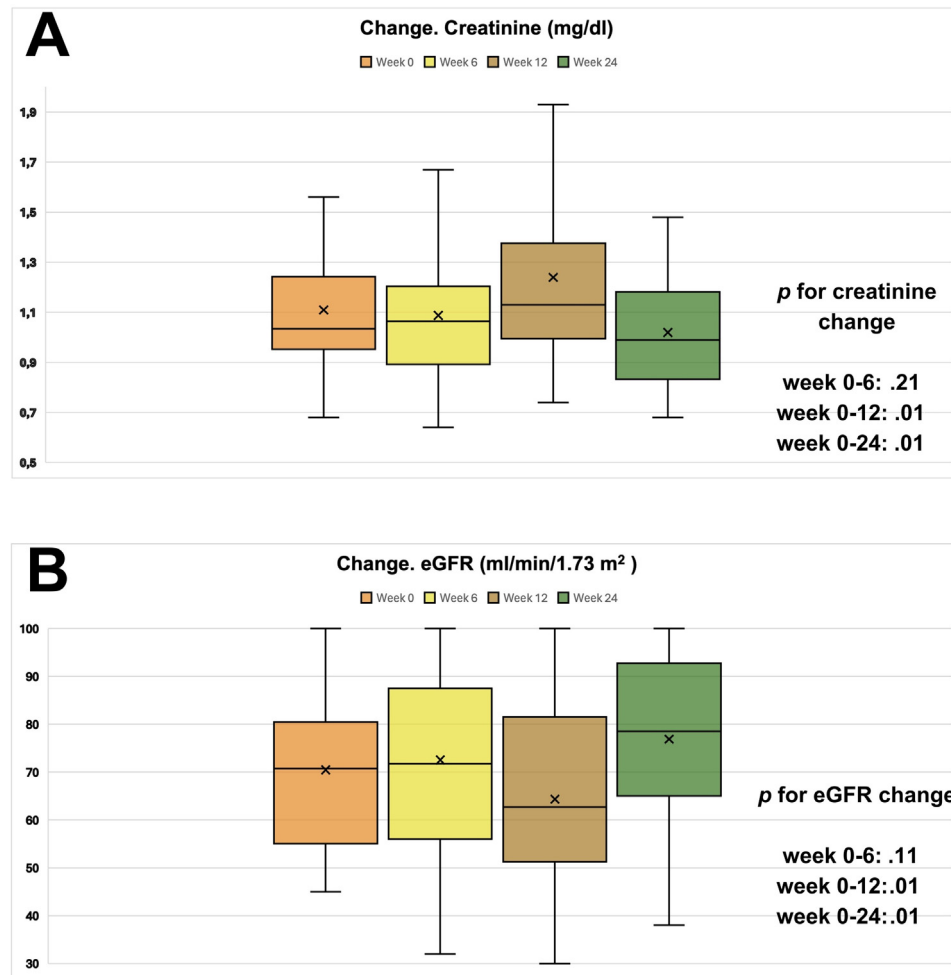


Fig. 5 – Changes in serum creatinine concentrations and estimated glomerular filtration rate at 6, 12 and 24 weeks compared to baseline values. Cr, creatinine; eGFR, estimated glomerular filtration rate.

parameter for evaluating exercise capacity and has a strong correlation with VO_2 peak.^{36,47} In addition, we did not estimate the sample size when designing the study, but a post hoc G*Power analysis showed a power of 1.00, confirming sufficient power to detect the large effect size for the primary outcome. Lastly, our non-controlled study design is susceptible to biases. The absence of a control group is justified as the primary aim was exploratory analysis to evaluate the potential effects of CM supplementation. Logistical constraints and limited resources made including a control group unfeasible. Future studies are planned to incorporate a control group, combine CM with a structured exercise regime and to measure VO_2 peak to provide more robust evidence on CM supplementation's clinical meaningful impact on patients with HF.

Conclusions

In conclusion, our study highlights the safety of CM supplementation in patients with HFrEF, showcasing significant enhancements in both functional capacity and HRQoL. These results suggest that CM may serve as a valuable complementary therapy for patients with HF.

What is known about the subject?

- CM is known to enhancing muscle strength and endurance in healthy individuals. Preliminary studies in patients with HFrEF suggest safety and a potential impact on functional capacity and HRQoL. However these studies are outdated, have small sample sizes and heterogeneous protocols.

Does it contribute anything new?

- Our study provides a much-needed update on the efficacy and safety of CM in HFrEF in the context of current treatment regimens, incorporating an extended follow-up period for safety evaluation. It confirms that CM supplementation is safe for patients with HFrEF and shows significant improvements in functional capacity and HRQoL. These findings suggest that CM could be a valuable adjunct therapy in the management of HFrEF.

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Ethical considerations

The protocol was approved by the hospital's Ethics Committee (Ref: E.O. 2021-69 SuplementacionCreatinalC-FEVIR; date of approval: 26/10/2021). Informed written consent was obtained from all participants. We confirm that SAGER guidelines regarding potential sex/gender bias have been followed.

Statement on the use of artificial intelligence

During the preparation of this work the authors acknowledge the use of ChatGPT as a supplementary tool to refine the manuscript's language, ensuring clarity and coherence while adhering to scientific writing standards. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Authors' contributions

J.C. López-Clemente: writing (review and editing, writing) original draft, visualization, validation, resources, project administration, methodology, investigation, formal analysis, data curation and conceptualization. L. Fuertes-Kenneally: writing (review and editing, writing) original draft, visualization, validation, project administration, formal analysis, data curation and conceptualization. J. Abellán Huerta: writing (review and editing), validation, investigation, data curation and conceptualization. F. Soria Arcos: writing (review and editing), validation, investigation, data curation and formal analysis. J. González-López: writing (review and editing), validation, investigation, data curation and methodology. R. Mármol Lozano: writing (review and editing), validation, investigation and data curation. J.A. Castillo Moreno: writing (review and editing), supervision, resources, project administration, methodology, investigation, formal analysis, data curation and conceptualization.

Conflicts of interest

The authors report no competing interests.

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